



How many colours do we need to colour the countries of a map in such a way that adjacent countries are coloured differently? How many days have to be scheduled for committee meetings of a parliament if every committee intends to meet for one day and some members of parliament serve on several committees? How can we find a school timetable of minimum total length, based on the information of how often each teacher has to teach each class?

A *vertex colouring* of a graph $G = (V, E)$ is a map $c: V \rightarrow S$ such that $c(v) \neq c(w)$ whenever v and w are adjacent. The elements of the set S are called the available *colours*. All that interests us about S is its size: typically, we shall be asking for the smallest integer k such that G has a k -colouring, a vertex colouring $c: V \rightarrow \{1, \dots, k\}$. This k is the (*vertex*-) *chromatic number* of G ; it is denoted by $\chi(G)$. A graph G with $\chi(G) = k$ is called k -chromatic; if $\chi(G) \leq k$, we call G k -colourable.

vertex
colouring

chromatic
number
 $\chi(G)$

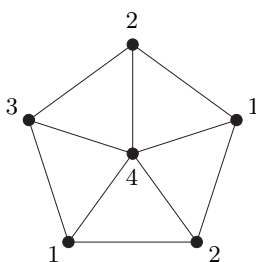


Fig. 5.0.1. A vertex colouring $V \rightarrow \{1, \dots, 4\}$

Note that a k -colouring is nothing but a vertex partition into k independent sets, now called *colour classes*; the non-trivial 2-colourable graphs, for example, are precisely the bipartite graphs. Historically, the colouring terminology comes from the map colouring problem stated

colour
classes

above, which leads to the problem of determining the maximum chromatic number of planar graphs. The committee scheduling problem, too, can be phrased as a vertex colouring problem – how?

edge
colouring

chromatic
index
 $\chi'(G)$

An *edge colouring* of $G = (V, E)$ is a map $c: E \rightarrow S$ with $c(e) \neq c(f)$ for any adjacent edges e, f . The smallest integer k for which a k -*edge-colouring* exists, i.e. an edge colouring $c: E \rightarrow \{1, \dots, k\}$, is the *edge-chromatic number*, or *chromatic index*, of G ; it is denoted by $\chi'(G)$. The third of our introductory questions can be modelled as an edge colouring problem in a bipartite multigraph (how?).

Clearly, every edge colouring of G is a vertex colouring of its line graph $L(G)$, and vice versa; in particular, $\chi'(G) = \chi(L(G))$. The problem of finding good edge colourings may thus be viewed as a restriction of the more general vertex colouring problem to this special class of graphs. As we shall see, this relationship between the two types of colouring problem is reflected by a marked difference in our knowledge about their solutions: while there are only very rough estimates for χ , its sister χ' always takes one of two values, either Δ or $\Delta + 1$.

5.1 Colouring maps and planar graphs

If any result in graph theory has a claim to be known to the world outside, it is the following *four colour theorem* (which implies that every map can be coloured with at most four colours):

Theorem 5.1.1. (Four Colour Theorem)

Every planar graph is 4-colourable.

Some remarks about the proof of the four colour theorem and its history can be found in the notes at the end of this chapter. Here, we prove the following weakening:

Proposition 5.1.2. (Five Colour Theorem)

Every planar graph is 5-colourable.

(4.1.1)
(4.2.10)

n, m

First proof. Let G be a plane graph with $n \geq 6$ vertices and m edges. We assume inductively that every plane graph with fewer than n vertices can be 5-coloured. By Corollary 4.2.10,

$$d(G) = 2m/n \leq 2(3n - 6)/n < 6;$$

v
 H
 c

let $v \in G$ be a vertex of degree at most 5. By the induction hypothesis, the graph $H := G - v$ has a vertex colouring $c: V(H) \rightarrow \{1, \dots, 5\}$. If c uses at most 4 colours for the neighbours of v , we can extend it to a 5-colouring of G . Let us assume, therefore, that v has exactly 5 neighbours, and that these have distinct colours.

Let D be an open disc around v , so small that it meets only those five straight edge segments of G that contain v . Let us enumerate these segments according to their cyclic position in D as $s_1, \dots, s_5$, and let vv_i be the edge containing s_i ($i = 1, \dots, 5$; Fig. 5.1.1). Without loss of generality we may assume that $c(v_i) = i$ for each i .

D
 $s_1, \dots, s_5$
 $v_1, \dots, v_5$

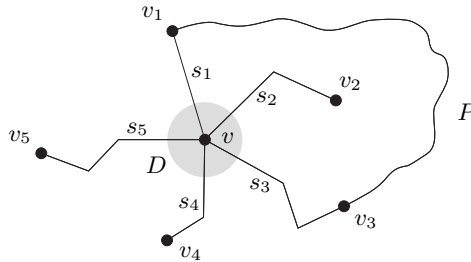


Fig. 5.1.1. The proof of the five colour theorem

Let us show first that every v_1 - v_3 path $P \subseteq H - \{v_2, v_4\}$ separates v_2 from v_4 in H . Clearly, this is the case if and only if the cycle $C := vv_1Pv_3v$ separates v_2 from v_4 in G . We prove this by showing that v_2 and v_4 lie in different faces of C .

P
 C

Let us pick an inner point x_2 of s_2 in D and an inner point x_4 of s_4 in D . Then in $D \setminus (s_1 \cup s_3) \subseteq \mathbb{R}^2 \setminus C$ every point can be linked by a polygonal arc to x_2 or to x_4 . This implies that x_2 and x_4 (and hence also v_2 and v_4) lie in different faces of C : otherwise D would meet only one of the two faces of C , which would contradict the fact that v lies on the frontier of both these faces (Theorem 4.1.1).

Given $i, j \in \{1, \dots, 5\}$, let $H_{i,j}$ be the subgraph of H induced by the vertices coloured i or j . We may assume that the component C_1 of $H_{1,3}$ containing v_1 also contains v_3 . Indeed, if we interchange the colours 1 and 3 at all the vertices of C_1 , we obtain another 5-colouring of H ; if $v_3 \notin C_1$, then v_1 and v_3 are both coloured 3 in this new colouring, and we may assign colour 1 to v . Thus, $H_{1,3}$ contains a v_1 - v_3 path P . As shown above, P separates v_2 from v_4 in H . Since $P \cap H_{2,4} = \emptyset$, this means that v_2 and v_4 lie in different components of $H_{2,4}$. In the component containing v_2 , we now interchange the colours 2 and 4, thus recolouring v_2 with colour 4. Now v no longer has a neighbour coloured 2, and we may give it this colour. $\square$

$H_{i,j}$

Second proof. As in the first proof, we assume inductively that every planar graph with fewer than $|G|$ vertices can be 5-coloured, and find a vertex v in G of degree 5. Since $K^5 \not\subseteq G$ by Corollary 4.2.11, this vertex v has non-adjacent neighbours u, w . Let $P = \{u, v, w\}$.

(4.2.10)
(4.2.11)
(4.4.6)

Since minors of planar graphs are again planar, e.g. by Kuratowski's theorem, G/P has a 5-colouring by the induction hypothesis. This in-

v

duces a 5-colouring of $G - v$ in which u and w receive the same colour, that of v_P from our colouring of G/P . The neighbours of v in this colouring of $G - v$ thus use at most four colours, and we can colour v with the fifth colour. $\square$

As a backdrop to the two famous theorems above, let us cite another well-known result:

Theorem 5.1.3. (Grötzsch 1959)

Every planar graph not containing a triangle is 3-colourable.

5.2 Colouring vertices

How do we determine the chromatic number of a given graph? How can we *find* a vertex-colouring with as few colours as possible? How does the chromatic number relate to other graph invariants, such as average degree, connectivity or girth?

Straight from the definition of the chromatic number we may derive the following upper bound:

Proposition 5.2.1. *Every graph G with m edges satisfies*

$$\chi(G) \leq \frac{1}{2} + \sqrt{2m + \frac{1}{4}}.$$

Proof. Let c be a vertex colouring of G with $k = \chi(G)$ colours. Then G has at least one edge between any two colour classes: if not, we could have used the same colour for both classes. Thus, $m \geq \frac{1}{2}k(k-1)$. Solving this inequality for k , we obtain the assertion claimed. $\square$

*greedy
algorithm*

One obvious way to colour a graph G with not too many colours is the following *greedy algorithm*: starting from a fixed vertex enumeration $v_1, \dots, v_n$ of G , we consider the vertices in turn and colour each v_i with the first available colour – e.g., with the smallest positive integer not already used to colour any neighbour of v_i among $v_1, \dots, v_{i-1}$. In this way, we never use more than $\Delta(G) + 1$ colours, even for unfavourable choices of the enumeration $v_1, \dots, v_n$. If G is complete or an odd cycle, then this is even best possible.

In general, though, this upper bound of $\Delta + 1$ is rather generous, even for greedy colourings. Indeed, when we come to colour the vertex v_i in the above algorithm, we only need a supply of $d_{G[v_1, \dots, v_i]}(v_i) + 1$ rather than $d_G(v_i) + 1$ colours to proceed; recall that, at this stage, the algorithm ignores any neighbours v_j of v_i with $j > i$. Hence in most graphs,